

Exam Date & Time: 18-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025 MATHEMATICAL FOUNDATIONS FOR DATA SCIENCE-I [MAT_2127]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) The two regression equations of the variables x and y are given below.
 $x = 19.13 - 0.87y$, $y = 11.64 - 0.50x$. Find
- A) (i) $\bar{x}$ and $\bar{y}$ (4)
 (ii) Correlation coefficient r
- B) Fit a parabola of the form $y = ax^2 + bx + c$ to the following data.
- | | | | | | |
|-----|-----|---|-----|---|---|
| x | 1 | 2 | 3 | 4 | 6 |
| y | 2.4 | 3 | 3.6 | 4 | 5 |
- (3)
- C) A person has undertaken a construction job. The probabilities are 0.65 that there will be strike, 0.80 that the construction job will be completed on time if there is no strike, and 0.32 that the construction job will be completed on time if there is a strike. Determine the probability that the construction job will be completed on time. (3)
- 2) Let X , Y and Z be uncorrelated random variables with standard deviation 5, 12, 9 respectively. Also, let $U = X+Y$ and $V = Y+Z$. Evaluate correlation coefficient ρ between U and V . (4)
- A)
- B) Suppose that the joint pdf of (X, Y) is given by
- $$f(x, y) = \begin{cases} e^{-y}, & \text{for } x > 0, \quad y > x \\ 0, & \text{elsewhere} \end{cases}$$
- (3)
- Evaluate conditional probability $P(X > 2 \mid Y < 4)$.
- C) (3)

Let X be a continuous random variable with probability density function $f(x)$ is defined by

$$f(x) = \begin{cases} ax, & 0 \leq x < 1 \\ a, & 1 < x < 2 \\ -ax + 3a, & 2 < x \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

Find

- (i) the value of 'a'
- (ii) the cumulative distribution function (c.d.f.) of X .

- 3) Two characteristics of a rocket engine's performance are thrust X and mixture ratio Y . Suppose that (X, Y) is a two-dimensional continuous random variable with joint pdf

$$f(x, y) = \begin{cases} 2(x + y - 2xy), & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases} \quad (4)$$

Find the marginal pdf of X and Y .

- B) In playing with an opponent of equal ability, which is more probable in each of the following? (3)
- i) Winning 2 games out of 4 OR 5 games out of 8?
 - ii) Winning at least 2 games out of 4 OR at least 5 games out of 8?
- C) Derive an expression for expectation of a continuous random variable having Normal distribution $(N(\mu, \sigma^2))$. (3)
- 4) If X_1 and X_2 are independent random variables and have standard normal distributions i.e. $X_1, X_2 \sim N(0, 1)$, find the pdf of $\frac{X_1}{X_2}$. (4)
- A)
- B) Show that sample variance is not an unbiased estimate of population variance. (3)
- C) Compute approximately the probability that the mean of the random sample of size 50 from a distribution having the pdf

$$f(x) = \begin{cases} 3x^2, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases} \quad (3)$$

is between $\frac{3}{5}$ and $\frac{4}{5}$.

- 5) (5)

- A) A tea company appoints four salesmen A, B, C, D and observes their sales in three seasons – Summer, Winter and Monsoons. The details are given below:

Salesman → Seasons ↓	A	B	C	D
Summer	36	36	21	35
Winter	28	29	31	32
Monsoon	26	28	29	29

- (i) Do the salesmen significantly differ in performance?
(ii) Is there significant difference between the seasons?

- B) Two researchers adopted different sampling techniques while investigating the same group of students to find the number of students falling in different intelligence levels. The results are as follows:

Researcher	Number of students in each level				Total
	Below avg.	Average	Above avg.	Genius	
X	86	60	44	10	200
Y	40	33	25	2	100
Total	126	93	69	12	300

(5)

Would you say that the sampling technique adopted by the two researchers are significantly different?

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Event Name	Subject
058-END SEMSTER-ODD-2025	MAT_2127:MATHEMATICAL FOUNDATIONS FOR DATA SCIENCE-I

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* NA - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025	MAT_2127:MATHEMATICAL FOUNDATIONS FOR DATA SCIENCE-I
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#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	14	50(100%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		14	50 (100%)